



# Parking Simulator Components

The **Car Parking Simulator** contains several components that work together to operate the parking facility, including **Barrier Gates**, **Parking Spots**, **Lights**, **Exhaust Fans**, and **Entry/Exit Spots**.

For example, when a car arrives at the parking entrance, the **Entry Spot** will notify you about the arriving car. You can then send a command to open the gate, guide the car to the best available parking spot, close the gate, and start counting the parking time. When the car wants to leave, it reaches the **Exit Spot**, which will notify you. You can then request the payment, verify it, and open the gate so the car can leave.

The Car Parking Simulator sends events to the **webhook URL** defined in your settings file in **JSON format**. In the other direction, your system communicates with the simulator through a **secure REST API** to send commands or make queries.

Components may break while operating, so you need to be aware of **predictive maintenance**, which should be performed before a component fails. Breakdowns can occur after a certain number of uses, such as with **Barrier Gates** and **Parking Spots**, or after a certain number of working hours, such as with **Exhaust Fans**.

If you perform maintenance before a component breaks, the maintenance will be **faster and cheaper**. Otherwise, repairs will be **slower and more expensive**. An important point to remember is that you **cannot operate a component while it is broken or under maintenance**, and attempting to do so will result in a penalty. You also cannot repair a component while it is in use—for example, you cannot repair a Parking Spot while a car is occupying it. Therefore, plan your maintenance schedule wisely.

Components are accessed by their **names**, while some components can also be accessed by **name or group**, such as Lights. Each component also belongs to a **Zone**, allowing you to organize and manage the parking structure more easily.

Server will send you notification when:

- Component is broken or
- Component is repaired
- Payment received
- Penalty is made
- Parking spot (car in out ) actions
- Entry/Exit gate action (car in/out) actions
- Carbon monoxide level change

## Component List:

### Car:

Cars are spawned automatically into the parking facility, and each car is assigned a unique plate number.

You can guide each car to a designated parking spot. If the car cannot reach the specified spot, or if the spot is already occupied, a penalty will be applied.

When a car reaches the exit, it will request a parking invoice. You must calculate the parking duration and applicable charge, then issue the invoice.

The car will then submit its payment. After validating the payment, guide the car through the exit and out of the parking facility.



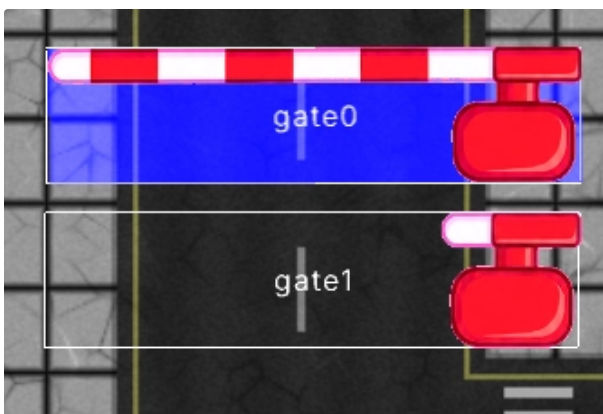
- Charging cost: 1 per each minute, multiply by 2 if electric

## Barrier Gate:

its main function is to block and let car pass the road, you can send commands to

`open` , `close` , `repair`

- Gates will notify you when it is fully open or fully closed .
- Gates will need to be repaired after X cycle of work (**Open/Close**)
- Barrier Gate require some time to open/close and you will be notified when fully open/fully close

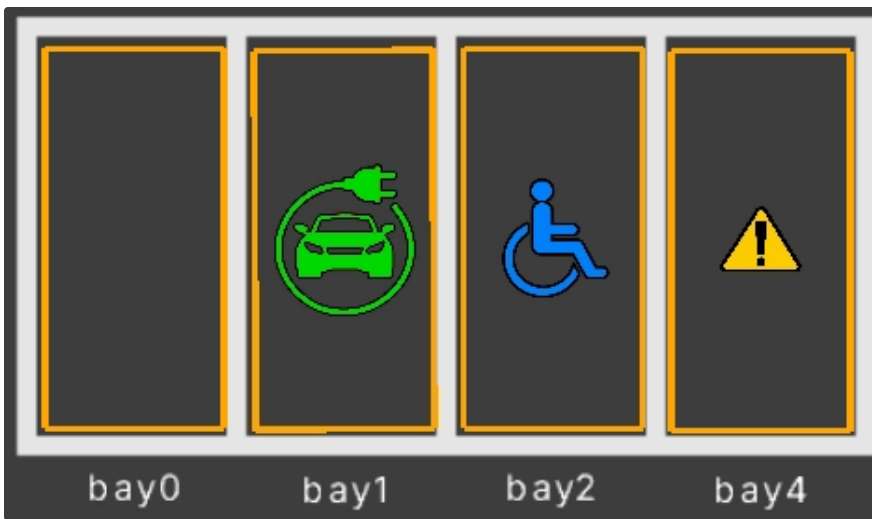


## Parking Spot:

used to provide a space for one car to park in, you can just send command to repair it

`repair`

- Each Parking spot will have name an Zone, and park type for (**any** , **electric**, **accessible**), make sure to send car to right spot type.
- Parking spot will send you signal when any car park in it or when it leave it, usually the sensor need less than a second to be stable and send the signal.
- Parking Spot will be broken after X time of usage
- `List`



## Exhaust Fan:

it works to reduce the carbon monoxide in its zone, it consume electricity, so better to turn off when Co levels are below 50 in a specific zone. You can send command to

`turn on` , `turn off` , `repair` the exhaust fan

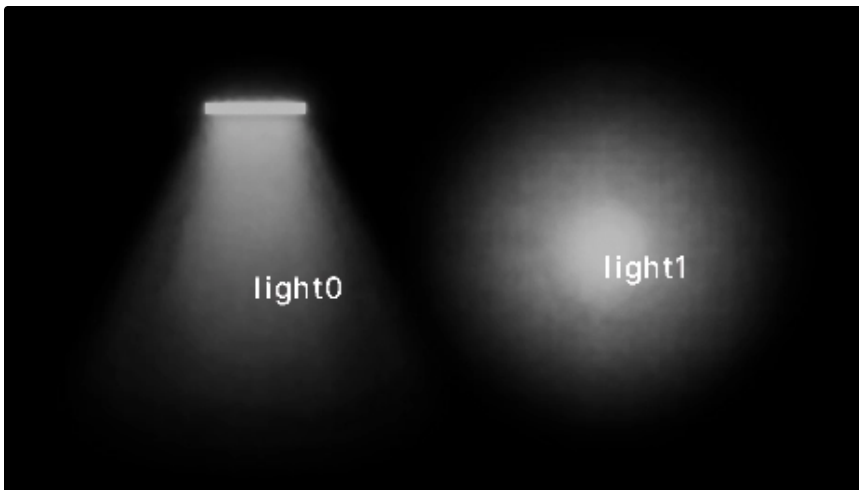
- Each exhaust fan will have a name and a zone
- it will be broken after X usage hour.
- Carbon monoxide is produced in zone only when car are moving



## Lights

lights are used to guide drivers in dark zones, and they consume electricity, since they are not modern tech, you can send signal to specific light to turn it on or off or to send command on or off to group of light at once.

- Please make sure they run only at night.
- No need to be on if all cars are parking, they must be on if car is moving in the zone



## Entry/Exit Spots:

Entry and Exit Spots represent sensors typically installed near barrier gates. Their purpose is to detect vehicles that are about to enter or leave the car park.

These sensors do not require any direct control. They only generate events containing information such as the car plate number and the detected action, for example **CarIn** or **CarOut** to that spot

## Entry Spots:

When a car arrives at an Entry Spot, you get an Webhook event provides information about the vehicle, including:

- Car plate number
- Car type
- Planned parking duration

Use this information to decide how the vehicle should be handled and where it should be directed.

- if car waited a lot it will leave.

## Exit Spots:

When a car reaches an Exit Spot, you can start asking for payment, verify it, then let car leave the park.

If payment has not yet been requested, request the (parking + electricity charging) payment **once**, Some cars will tweak the system and send fake payment.

- parking cost = total minutes spent parking, multiplied by 2 if car is electric
- If the car waited a lot it may run away

## Broken Component:

Components may break due to over usage, or long running time.

if component is broken you will see this icon on it



if component is under maintenance you will see this wrench icon with green part of it represent remaining time to finish:

